Lorenzo **Villanueva**

PhD Research Engineer with 6+ years of experience in high-speed instrumentation and FPGA control. Expert in hardware-software co-design for feedback loops, with a background in Quantum Nano Devices

Field Application Specialist | Lab Automation & Scientific Instrumentation · PhD Biotechnology · Data Scientist

Laboratory of Dynamics and nanoenvironment of biological membranes (DyNaMo), Marseille, FR

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Employment

Dynamics and nanoenvironment of biological membranes (DyNaMo) Laboratory — *Marseille, France*

PhD - Universite d'AIX Marseille - Ecole Doctorale Science du vivant (ED658) — *Jun. 2022 - May. 2026*

- Developed and automated HS-AFM data acquisition workflows, including custom FPGA control software and Python-based post-analysis pipelines for biological force measurements
- Hired by ERC funded project (MechaDynA)(OpenFMLab)

Adhesion et Inflammation (LAI) Laboratory — *Marseille, France*

INSERM TEST & LAB AUTOMATION Engineer — *Mar. 2019 - Jun. 2022*

- Developed LabVIEW and NI FPGA software for high-speed data acquisition for High Speed atomic force microscopy (HSAFM)
- Implemented control algorithms for long range HSAFM for data density and total acqusition time. Developed in LabVIEW FPGA
- Hands on experience in electronics such as photodiode transimepdance amplifiers, high power
- ERC funded project (MechaDynA)

Integrated Micro-electronics, Inc. (IMI) — *Biñan Laguna, Philippines*

Test-Development Engineer — *Nov. 2016 - Jan. 2019*

- Developed LabVIEW-based automated test software and hardware for production, created test system documentation, supported overseas deployment, troubleshot test issues, and optimized production processes for efficiency.

Shell Philippines Exploration B.V (SPEX) — *Manila, Philippines*

Electrical Engineer Intern — *Jan. - Sept. 2016*

- Conducted a feasibility study on the CCGT (Combines Cycle Gas Turbine Engine) in Malampaya Offshore Gas plant to analyzed the options in either to utilize for Gas plant or for the allocation of electricity to sell to the National grid

Education

Universite d'AIX Marseille — *Marseille, France*

PhD Biotechnology — *2024 – 2026*

- Long-range high-speed force spectroscopy applied to biophysics
- Supervisors: Felix Rico, Co-supervisor Qingze Zou

Universite d'AIX Marseille — *Marseille, France*

Master in Nanosciences et Nanotechnologies (Quantum Nano Devices) — *2020 – 2023*

- M2: Design and Implementation of a Force Control algorithm in LabVIEW FPGA for high-speed atomic force microscopy (HSAFM)
- Supervisors: Dr. Felix Rico and Dr. Jorge Rodriguez-ramos
- M1: Construction and validation of high-speed piezo positioner with endcups for HSAFM using a homebuilt Michealson Interferometric sensor
- Supervisors: Dr. Felix Rico and Dr. Jorge Rodriguez-ramos

Purdue University — *West Lafayette, Indiana, USA*

BS Electrical Computer Engineering Technology (ECET) — *2012 – 2016*

- Analog I Lab Assistant (Aug. - Dec. 2014): Troubleshooting digital and analog circuits, collaborating with teaching assistants, working with Arduino for photodiode switching, MOSFET control, and conducting lab safety protocols.
- Thesis Project: Lightning Detection System located in indigenous region Quehue, Peru
- Study Abroad (Jan. - May 2016) - Part-time scholarship in Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru

Work-related Experience

DyNaMo, AIX Marseille Universite

Marseille, France

labVIEW SW & HW Developer, PhD candidate — *Jan. 2022 – current*

- **Interdisciplinary**: Collaborated directly with experimental researchers to define and optimize force-indentation data workflows, translating complex hardware requirements into user-friendly acquisition systems
- **FPGA SW Development** (Aug. 2023 – current): Designed and maintained custom modules for HS-AFM system. Optimized DMA transfers for high-speed data acquisition, Managed high-bandwidth data streams between FPGA and host via PCIe, implemented DRAM buffering to output waveforms [labVIEW]
- **NeoLaura - HS-AFM SW** (Jan. 2019 – Aug. 2022): Upgraded and revised a new user interface for force spectroscopy [labVIEW]
- **Jetson NANO C100 homeamde AFM** (Jan. 2021 – Aug. 2022):Setup a linux system on Jetson NANO C100 for a homemade AFM system with redpitaya with LAN connection for remote control and data acquisition [Linux, VHDL]
- **OpenFMLab** (Jan. 2022 – current): Supervising master student for designing 3D printed XY stage scanner with mechanical gains for range increase
- **PyFMlab** (Jan. 2022 – current): Adapted and maintained a Python-based software for controlling the AFM system, including data acquisition, analysis, and visualization

LAI, AIX Marseille Universite

Marseille, France

labVIEW SW & HW Developer, Research Engineer — *Jan. 2019 – Jan. 2022*

- **Laura - HS-AFM SW** (Jan. 2019 – Aug. 2022): Upgraded and revised an old user interface for force spectroscopy, coupled to Ando's HS-AFM system (RIBM) [labVIEW]. Implemented retract-chirp method, Power spectrum analysis, and optimized data saving
- **Technician** (Jan. 2022 – current): Repaired and replaced Z piezos for HS AFM scanners for imagine.
- Data analysis of AFM data using Python.

Integrated Micro-electronics, Inc. (IMI)

Biñan, Laguna, Philippines

labVIEW SW Dev & Test System Dev Engr. — *Jan. 2016 – Aug. 2018*

- **Sensor Handler Integration for AC Withstanding Tester** (Jun. – Aug. 2018): Supported 2 projects in developing customized GUI interface using GPIB and RS232 protocol to sensor and communicate with third party Withstanding Voltage Tester (TDK outfit) system executed in LabVIEW with SQL database integration
- **HTGB/HTRB/H3TRB IGBT Tester System Development** (Aug. 2017 – Aug. 2018): Devised a LabVIEW GUI for sub-assembly handler integration, implemented HT/GB and HTTRB testing for power module validation, added SQL-based traceability, and designed a custom testing box for measurements
- **DUT Controller Development** (Dec. 2017 – Aug. 2018): Developed Outlet IoT Scheduler device test system for debugging and troubleshooting hardware via CAN protocol, and integrated IoT tasks using Zapier module orchestrated with JIRA
- **Hardware Drivers Development** (Jan. 2016 – May 2017): Traveled to IMI, Jiaxing, China to support Test System Development team in developing Oscilloscope drivers in LabVIEW, LVOOP method for dynamic dispatch of test equipment to replace SQL database ensuring data integrity
- **Stray-light Optical Tester** (Nov. 2016 – Mar. 2017): Reviewed the system architecture of GUI and SW Architecture in LabVIEW for Optical Stray-light Tester and hardware design using Prindle Optical Theory; designed a PID controller to drive Verily Optical Tester Hardware for pneumatic movement and sensor alignment

Peer review publications

First-author publications:

1 PREDICTION-BASED ALGORITHMS FOR LONG-RANGE HIGH-SPEED FORCE SPECTROSCOPY *in prep.*

Villanueva, L., *Saravanan, Y., Eroles, M., Couderc, A., Rodriguez-Ramos, J., Zou, Q. & Rico, F.* *Journal of Molecular Recognition*

Second-author publications:

2 NEAR EQUILIBRIUM UNBINDING OF STREPTAVIDIN/BIOTIN USING SINGLE MOLECULE ACOUSTIC FORCE SPECTROSCOPY *2025*

Saravanan, Y., ***Villanueva, L.***, *Leveque, C., Modesti, M., El, O., Valotteau, C. & Rico, F.* *NanoLetters*

N-author publications:

3 COMBINING DNA SCAFFOLDS AND ACOUSTIC FORCE SPECTROSCOPY TO CHARACTERIZE INDIVIDUAL PROTEIN BONDS *2023*

Wang, Y.J., Valotteau, C., Aimard, A., ***Villanueva, L.***, *Kostrz, D., Follenfant, M., Strick, T., Chames, P., Rico, F., Gosse, C. & Limozin, L.* *Biophysical Journal*

Seminars & Conference Talks

CENTURI - Hackathon 2025 — *Marseille, France*
INTERDISCIPLINARY SCIENTIFIC COLLABORATION AND INNOVATION WORKSHOP — *Jun. 20-22, 2025*

AFM Biomed Conference 2025 — *Barcelona, Spain*
INTERNATIONAL CONFERENCE ON ATOMIC FORCE MICROSCOPY APPLICATIONS IN BIOMEDICINE — *Apr. 08-11, 2025*
Flash Talk: Control Algorithms for Long-Range High-Speed AFM

CENTURI - Data Week Part 2 — *Marseille, France*
ADVANCED DATA ANALYSIS AND COMPUTATIONAL METHODS IN LIFE SCIENCES — *Mar. 03-07, 2025*

CENTURI - Statistics (by Pierre Mangeol) — *Marseille, France*
STATISTICAL METHODS AND ANALYSIS FOR BIOLOGICAL RESEARCH — *Feb. 24-28, 2025*

CENTURI - Hackathon 2024 — *Marseille, France*
TEAM STREAMQUEST: AUTOMATED STEM CELL QUALITY CONTROL WITH AUTOMATION | VIRGILE VIAUNOFF - DEVELOPED AN AI-DRIVEN SMART CAMERA SYSTEM INTEGRATING MICROSCOPY AND ROBOTICS FOR AUTOMATED STEM CELL QUALITY ASSESSMENT AND DIFFERENTIATION — *Jun. 28-30, 2024*

Cell Culture & Good Practices in Cell Culture — *Marseille, France*
PROFESSIONAL TRAINING IN CELL CULTURE TECHNIQUES AND LABORATORY BEST PRACTICES — *Apr. 03, 2024*

Python Scripts in ImageJ — *Marseille, France*
ADVANCED IMAGE ANALYSIS AND AUTOMATION USING PYTHON IN IMAGEJ — *Jan. 24 - Feb. 13, 2024*

CENTURI - Hackathon 2023 (1 out of 2 winning teams) — *Marseille, France*
TEAM BALLERINA: AUTOMATED MOTION CORRECTION FOR MINIMALLY INVASIVE IMAGING DEVELOPING TISSUES | ERIC LOGOFF — *Jun. 14-16, 2023*

Training

Summer schools

Jun. 02-07, 2024	**Mecabio - CNRS Thematic School**, Quantitative and Predictive Approaches in Biomechanics and Mechanobiology for Health	*Grenoble, France*
Jun. 2022	NANOSUM, International summer school on Nanosciences & Nanotechnologies - Poster	*La Ciotat, France*
Jul. 2022	**13th AFMBioMed**, Introduction to Atomic Force Microscopy in Life Sciences and Medicine	*Marseille, France*
Jul. 2019	**11th AFMBioMed**, Introduction to Atomic Force Microscopy in Life Sciences and Medicine with hands-on training using JPK Instruments cell Vega system	*Grenoble, France*

Field works

Academic collaboration with University of Barcelona — *Barcelona, Spain*
Gavaralab - CYTOSKELETAL MECHANICS GROUP, FACULTY OF MEDICINE AND HEALTH SCIENCES — *Mar. - Apr. 2025*

- Designed top-bottom spot 4DMI photodiode circuit using KiCad
- Implemented HS-AFM software NeoLaura with their hardware (ongoing)

Leadership & Community Service

PyFMLab GitHub Repository

CONTRIBUTOR

- Developed and maintained a Python library to read and write TDMS files, facilitating data exchange between LabVIEW and Python routines
- Collaborated with the open-source community to enhance the library's functionality and usability.

Skills

Programming	Python, Fortran, C, C++,LabVIEW, LabVIEW FPGA, Matlab, Bash, $\TeX$
Data analysis	HSAFM data processing, ImageJ/Fiji, Gwyddion
3D Printing & CAD	Fusion 360
Electronics	PCB design (kiCad), Circuit design, Soldering, Instrumentation
Video	DaVinci Resolve

Activities

Extra - Activities

- Centuri Hackathon 2026 Team Leader, openAFM project: Developed an open-source software on Gwyscope controller
- MUST13 - Football association supporting LGBTQ equality rights within the sport community (Mar. 2024 - current)
- Futsal Club Founder - (IMI)(May 2018 - Aug. 2018)
- Volunteer, One Million Lights (OML), Philippines: Provide clean, safe, affordable solar-powered LED lamps in rural communities (Aug. 2011 - Jan. 2012)
- Vice Chairman, HS Student Council (SY 2011 - 2012)

Hobbies	Futsal, Soccer, Baseball, Long-boarding, Badminton, Frisbee, Guitar, Piano, Digital Synthesizers, Swimming, Hiking, Biking
Travels	Philippines, USA, China, Canada, France, Peru
Interests	Building and designing products for technology and innovation

Other

Citizenship	Filipino
Birth year	1994 (31 years old)
Languages	English (native), Tagalog (native), French (B1), Spanish(A2)
Driving	French B/European licence